Coding Basics ⌨

Junior Coding for Good Badge — Girls Build Bots

Presented by your robotics mentors!

Coding *for good* 💚

Programmers build things that **solve problems for real users**.

Today, in every activity, we're going to keep asking:

Who is this for?

Algorithm Relay 🃏

In teams, write an algorithm on cards for a physical task: - Stack three cups into a tower 🥤 - Do a 4-move dance 💃

Then another team runs it **exactly as written**.

No interpreting. No guessing. 😈

What just happened? 🤔

Missing steps → the task breaks. Wrong order → the task breaks.

Computers do exactly what we say.

So precision is the coder’s job.

Loops save work 🔁

Without a loop: > clap, clap, clap, clap, clap, clap

With a loop: > **repeat 6:** clap

Same result. Way less writing.

Where are the loops hiding in real life? *Brushing every tooth · dealing every card*

Today’s three big ideas 🧰

Idea	What it does
Sequence	steps in order
Event	when THIS happens, do THAT
Loop	repeat without rewriting

Everything you build today uses these three.

Build: one concept at a time 💻

In MakeCode Arcade:

1. **Sequence** — a sprite appears and walks across the screen
2. **Event** — it moves when you press the arrow keys
3. **Loop** — a star respawns *forever* after you collect it
4. **Challenge** — add a counter for how many stars you've collected

Finished early? Become a **deputy mentor** and help a neighbor! 🎓

Debug Hunt 🐛

Your pair gets a **broken program**.

1. Find the bug
2. Fix it
3. Explain **what was wrong**

Real programmers spend more time debugging than writing.

Robot moment 🤖

Our robot code uses the **exact same three ideas**:

- **Sequences** → autonomous routines
- **Events** → driver buttons
- **Loops** → checking sensors

Just in a bigger language.

Closing circle 🎯

Which concept would you teach a friend first?

Sequence · Loop · Event

What you accomplished ✅

- Wrote an algorithm precise enough for anyone to follow
- Explained what a loop does and why coders use them
- Built a program with a sequence, an event, and a loop
- Found a bug, fixed it, and explained what was wrong

Keep going at home! 🏠

- **arcade.makecode.com** — keep building (free)
- **scratch.mit.edu** — games, stories, animations
- Ask your troop leader about the next Coding for Good badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.